7.6 The optical absorption cross-section

Monochromatic radiation causes an atom to undergo Rabi oscillations, but when the transition has damping the atom settles down to a steady state in which the excitation rate equals the decay rate. This has been shown explicitly above for an optical transition with spontaneous emission, but the same reduction of the coherent evolution of quantum amplitudes to a simple rate equation for populations (amplitudes squared) also occurs for other line-broadening mechanisms, e.g. Doppler broadening (Chapter 8) and collisions. Thus the equilibrium situation for monochromatic radiation is described by rate equations like those in Einstein's treatment of excitation by broadband radiation (eqns 1.25). It is convenient to write these rate equations in terms of an optical absorption cross-section defined in the usual way, as in Fig. 7.4. Consider a beam of particles (in this case photons) passing through a medium with N atoms per unit volume.[38] A slab of thickness Δz has $N\Delta z$ atoms per unit area and the fraction of particles absorbed by the target atoms is $N\sigma\Delta z$, where σ is defined as the cross-section; $N\sigma\Delta z$ gives the fraction of the target area covered by the atoms and this equals the probability that an incident particle hits an atom in the target (as it passes through the slab). The parameter σ that characterises the probability of absorption is equally well definable in quantum mechanics (in which photons and particles are delocalised, fuzzy objects) even though this cross-section generally has little relation to the physical size of the object (as we shall see). The probability of absorption equals the fraction of intensity lost, $\Delta I/I = -N\sigma\Delta z$, so the attenuation of the beam is described by

$$\frac{\mathrm{d}I}{\mathrm{d}z} = -\kappa(\omega)\, I = -N\sigma(\omega) I\,, \tag{7.70}$$

where $\kappa(\omega)$ is the absorption coefficient at the angular frequency ω of the incident photons. Integration gives an exponential decrease of the intensity with distance, namely

$$I(\omega, z) = I(\omega, 0) \exp\{-\kappa(\omega)\, z\}\,. \tag{7.71}$$

[38] N represents the number density and has dimensions of m^{-3}, following the usual convention in laser physics.

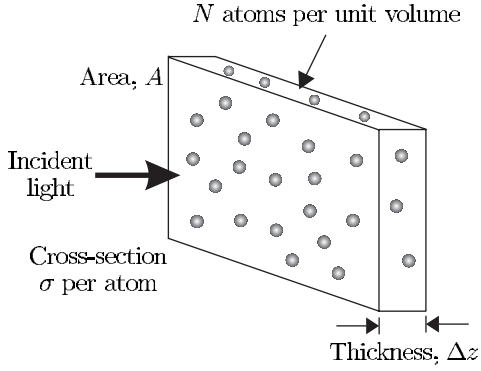


Fig. 7.4 Atoms with number density N distributed in a slab of thickness Δz absorb a fraction $N\sigma\Delta z$ of the incident beam intensity, where σ is defined as the cross-section (for absorption). $N\Delta z$ is the number of atoms per unit area and σ represents the 'target' area that each atom presents. We assume that the motion of the target atoms can be ignored (Doppler broadening is treated in Exercise 7.9) and also that atoms in the next layer (of thickness δz) cannot 'hide' behind these atoms (see Brooker 2003, Problem 3.26).

This formula, known as Beer's law (see Fox 2001), works well for absorption of low-intensity light that leaves most of the population in the ground state. Intense laser light significantly affects the populations of the atomic levels and we must take this into account. Atoms in level 2 undergo stimulated emission and this process leads to a gain in intensity (amplification) that offsets some of the absorption. Equation 7.70 must be modified to[39]

$$\frac{\mathrm{d}I}{\mathrm{d}z} = -\kappa(\omega) I(\omega) = -(N_1 - N_2)\sigma(\omega) I(\omega). \tag{7.72}$$

Absorption and stimulated emission have the same cross-section. For the specific case of a two-level atom this can be seen from the symmetry with respect to the exchange of the labels 1 and 2 in the treatment of the two-level atom in the early parts of this chapter; the oscillating electric field drives the transition from 1 to 2 at the same rate as the reverse process—only the spontaneous emission goes one way. This is an example of the general principle that a strong absorber is also a strong emitter.[40] This is also linked to the equality of the Einstein coefficients, $B_{12} = B_{21}$, for non-degenerate levels. The population densities in the two levels obey the conservation equation $N = N_1 + N_2$.[41] In the steady state conservation of energy per unit volume of the absorber requires that

$$(N_1 - N_2)\sigma(\omega) I(\omega) = N_2 A_{21} \hbar\omega. \tag{7.73}$$

On the left-hand side is the amount by which the rate of absorption of energy exceeds the stimulated emission, i.e. the net rate of energy absorbed per unit volume. On the right-hand side is the rate at which the atoms scatter energy out of the beam—the rate of spontaneous emission for atoms in the excited state times $\hbar\omega$.[42] The number densities are related to the variables in the optical Bloch equations by $\rho_{22} = N_2/N$ and

$$w = \frac{N_2 - N_1}{N}, \tag{7.74}$$

and w and ρ_{22} are given in eqns 7.68 and 7.69, respectively. Hence

$$\sigma(\omega) = \frac{\rho_{22}}{w}\frac{A_{21}\hbar\omega}{I} = \frac{\Omega^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4} \times \frac{A_{21}\hbar\omega}{I}. \tag{7.75}$$

Both I and Ω^2 are proportional to $|E_0|^2$ so this cancels out, and further manipulation yields[43]

$$\sigma(\omega) = 3 \times \frac{\pi^2 c^2}{\omega_0^2} A_{21}\, g_{\mathrm{H}}(\omega). \tag{7.76}$$

The Lorentzian frequency dependence is expressed by the line shape function

$$g_{\mathrm{H}}(\omega) = \frac{1}{2\pi}\frac{\Gamma}{(\omega - \omega_0)^2 + \Gamma^2/4}, \tag{7.77}$$

where the subscript H denotes homogeneous, i.e. something that is the same for each atom, like the radiative broadening considered here.[44] The

[39] We do not try to include the degeneracy of the levels because illumination with intense polarized laser light usually leads to *unequal* populations of the states with different M_J, or M_F. This differs from the usual situation in laser physics where the excitation, or pumping mechanisms, populate all states in a given level at the same rate, so N_1/g_1 and N_2/g_2 can be taken as the population densities per state. Selective excitation of the upper level can give $N_2/g_2 > N_1/g_1$, and hence gain.

[40] The laws of thermodynamics require that an object stays in equilibrium with black-body radiation at the same temperature, hence the absorbed and emitted powers must balance.

[41] Compare this with eqns 1.26, 7.7 and 7.43.

[42] This assumes that atoms do not get rid of their energy in any other way such as inelastic collisions.

[43] Intensity is related to the electric field amplitude by $I = \epsilon_0 c |E_0(\omega)|^2/2$, and $\Omega^2 = e^2 X_{12}^2 |E_0|^2/\hbar^2$ (eqn 7.12). Also $X_{12}^2 = |D_{12}|^2/3 \propto A_{21}$ (eqn 7.23). The degeneracy factors are $g_1 = g_2 = 1$ for the two-level atom, but see eqn 7.79.

[44] It is a general result that homogeneous broadening mechanisms give a Lorentzian line shape.

area under this line shape function equals unity:

$$\int_{-\infty}^{\infty} g_{\mathrm{H}}(\omega)\,\mathrm{d}\omega = 1\,. \tag{7.78}$$

The pre-factor of 3 in eqn 7.76 may have any value in the range 0 to 3. It has the maximum value of 3 for atoms with the optimum orientation to absorb a beam of polarized laser light (from a specific direction). However, if either the light is unpolarized or the atoms have a random orientation (i.e. they are uniformly distributed across all the M_J states or M_F states) then the pre-factor is 1 because $|X_{12}|^2 = |D_{12}|^2/3$ as in eqn 7.21 (from the average of $\cos^2\theta$ over all angles), and this 1/3 cancels the pre-factor of 3.[45] Under these conditions the absorption does not depend on the magnetic state (M_J or M_F) so a real atom with degenerate levels has a cross-section of

$$\sigma(\omega) = \frac{g_2}{g_1} \times \frac{\pi^2 c^2}{\omega_0^2} A_{21}\; g_{\mathrm{H}}(\omega)\,. \tag{7.79}$$

This equation, or eqn 7.76, applies to many experimental situations. Careful study of the following examples gives physical insight that can be applied to other situations.[46]

[45] Spontaneously emitted photons go in random directions so an average over angles always occurs in the calculation of A_{21}.

[46] Typically, for atoms with a well-defined orientation the polarization of the light is chosen to give the maximum cross-section. If this is not the case then the angular momentum algebra may be required to calculate the matrix elements. Only in special cases would the polarization be chosen to give a very weak interaction, i.e. a pre-factor much less than unity in eqn 7.76.

Example 7.2 *Atoms in a specific M_F state interacting with a polarized laser beam, e.g. sodium atoms in a magnetic trap that absorb a circularly-polarized probe beam (Fig. 7.5)*
This gives effectively a two-level system and the polarization of the light matches the atom's orientation so eqn 7.76 applies (the pre-factor has the maximum value of 3). To drive the $\Delta M_F = +1$ transition the

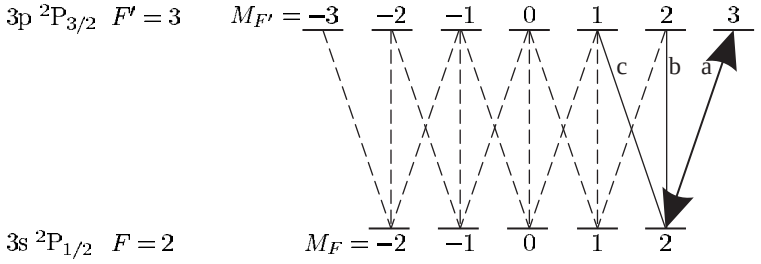


Fig. 7.5 The Zeeman states of the 3s $^2\mathrm{P}_{1/2}$ $F = 2$ and 3p $^2\mathrm{P}_{3/2}$ $F' = 3$ hyperfine levels of sodium, and the allowed electric dipole transitions between them. The other hyperfine levels ($F = 1$ and $F' = 0, 1$ and 2) have not been shown. Excitation of the transition $F = 2,\ M_F = 2$ to $F' = 3,\ M_{F'} = 3$ (labelled a) gives a closed cycle that has similar properties to a two-level atom—the selection rules dictate that atoms in the $F' = 3,\ M_{F'} = 3$ state spontaneously decay back to the initial state. (Circularly-polarized light that excites $\Delta M_F = +1$ transitions leads to cycles of absorption and emission that tend to drive the population in the $F = 2$ level towards the state of maximum M_F, and this optical pumping process provides a way of preparing a sample of atoms in this state.) When all the atoms have the correct orientation, i.e. they are in the $F = 2,\ M_F = 2$ state for this example, then eqn 7.76 applies. Atoms in this state give less absorption for linearly-polarized light (transition b), or circular polarization of the wrong handedness (transition c).

circularly-polarized light must have the correct handedness and propagate along the atom's quantisation axis (defined by the magnetic field in this example).[47]

[47] The direction of the electric field at the atom depends on both the polarization *and* direction of the radiation, e.g. circularly-polarized light that propagates perpendicular to the quantisation axis drives $\Delta M_F = 0$ and ± 1 (π- and σ-) transitions. This leads to a smaller cross-section than when the light propagates along the axis. Radiation that propagates in all directions does not produce a polarized electric field, e.g. isotropic radiation in a blackbody enclosure.

Example 7.3 *The absorption of light on an* s–p *transition, e.g. the* 3s–3p *resonance line of sodium*
Light with a particular polarization and direction drives a transition to one magnetic sub-level in the upper level, as shown in Fig. 7.6(a). Since the lower level has only $m_l = 0$ there is no averaging over the orientation and eqn 7.76 applies. Unpolarized light drives transitions to the three upper m_l states equally. For each transition the averaging gives a factor of 1/3 but all three transitions contribute equally to the absorption so the atoms have the same cross-section as for polarized light (eqn 7.79 with $g_2/g_1 = 3$). Thus the s–p transition is a special case that gives the same absorption cross-section whatever the polarization of the light. Atoms with $m_l = 0$ have no preferred direction and interact in the same way with light of any polarization (or direction). In contrast, for the p–s transition shown in Fig. 7.6(b), atoms in a given m_l state only interact with light that has the correct polarization to drive the transition to $m_l = 0$.

7.6.1 Cross-section for pure radiative broadening

The peak absorption cross-section given by eqn 7.76, when $\omega = \omega_0$, is

$$\sigma(\omega_0) = 3 \times \frac{2\pi c^2}{\omega_0^2} \frac{A_{21}}{\Gamma}. \tag{7.80}$$

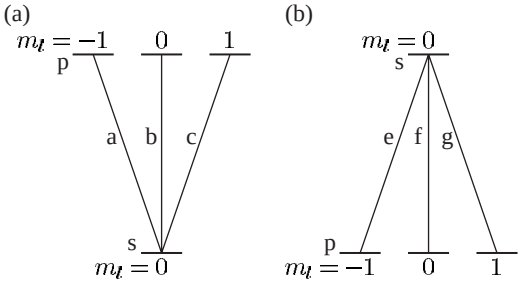


Fig. 7.6 A comparison of s–p and p–s transitions. (a) The three transitions a, b and c between the s and p levels have equal strength. The physical reason for this is that the spontaneous decay rate of the upper m_l states cannot depend on the atom's orientation in space. Light linearly-polarized parallel to the z-axis drives π-transition b only, and spontaneous decay occurs back to the initial state since there are no other accessible states—this gives the equivalent of a two-level system. The s–p transition is a special case where the absorption does not depend on the polarization, e.g. unpolarized light gives equal excitation rates on the three transitions a, b and c, and this increases the absorption by the degeneracy factor $g_2/g_1 = 3$, thereby cancelling the 1/3 that arises in the orientational average. (b) In contrast, for the p–s transition the peak cross-section is one-ninth of that in (a).[48]

[48] Spin is ignored here. This applies when either the fine structure is not resolved, e.g. this may arise for the transition 2s–3p in hydrogen where the fine structure of the upper level is small, or to transitions between singlet terms, i.e. ^{1}S–^{1}P and ^{1}P–^{1}S (with $m_l \rightarrow M_l$ in the figure).

In a two-level atom the upper level can only decay to level 1 so $\Gamma = A_{21}$, and for a transition of wavelength $\lambda_0 = 2\pi c/\omega_0$ we find

$$\sigma(\omega_0) = 3 \times \frac{\lambda_0^2}{2\pi} \simeq \frac{\lambda_0^2}{2}. \tag{7.81}$$

This maximum cross-section is much larger than the size of the atom, e.g. the $\lambda_0 = 589\,\text{nm}$ transition of sodium has $\sigma(\omega_0) = 2 \times 10^{-13}\,\text{m}^2$, whereas in kinetic theory the atoms have a cross-section of only $\pi d^2 = 3 \times 10^{-18}\,\text{m}^{-2}$ for an atomic diameter of $d = 0.3\,\text{nm}$—'collisions' between atoms and photons have a large resonant enhancement. The optical cross-section decreases rapidly off resonance, e.g. light of wavelength 600 nm gives $\Gamma/(\omega - \omega_0) = 10^{-6}$ for the sodium transition above, so that $\sigma(\omega) = 10^{-12} \times \sigma(\omega_0) = 2 \times 10^{-25}\,\text{m}^2$. Clearly the absorption of radiation has little relation to the size of the electronic orbitals.

7.6.2 The saturation intensity

In the previous section we calculated the absorption cross-section starting from eqn 7.73 and we shall now use the same equation to determine the population difference; we can write eqn 7.73 as $(N_1 - N_2) \times r = N_2$, where the dimensionless ratio $r = \sigma(\omega) I(\omega)/(\hbar\omega A_{21})$. This equation and $N_1 + N_2 = N$ give the difference in population densities as

$$N_1 - N_2 = \frac{N}{1 + 2r} = \frac{N}{1 + I/I_\text{s}(\omega)}, \tag{7.82}$$

where the saturation intensity is defined by

$$I_\text{s}(\omega) = \frac{\hbar\omega A_{21}}{2\sigma(\omega)}. \tag{7.83}$$

It is important to note that other definitions of saturation intensity are also used, such as the above expression without 2 in the denominator. From eqn 7.72 we find that the absorption coefficient depends on intensity as follows:[49]

$$\kappa(\omega, I) = \frac{N\sigma(\omega)}{1 + I/I_\text{s}(\omega)}. \tag{7.84}$$

The *minimum* value of $I_\text{sat}(\omega)$ occurs on resonance where the cross-section is largest; this minimum value is often referred to as *the* saturation intensity, $I_\text{sat} \equiv I_\text{s}(\omega_0)$, given by[50]

$$I_\text{sat} = \frac{\pi}{3}\frac{hc}{\lambda^3\tau}, \tag{7.85}$$

where $\tau = \Gamma^{-1}$ is the lifetime for radiative broadening. For example, the resonance transition in sodium at $\lambda = 589\,\text{nm}$ has a lifetime of $\tau = 16\,\text{ns}$ and for an appropriate polarization (as in Fig. 7.5) the atom cycles on an effectively two-level transition. This leads to an intensity $I_\text{sat} = 60\,\text{W m}^{-2}$, or $6\,\text{mW cm}^{-2}$, that can easily be produced by a tunable dye laser.

[49] This equation is very similar to the formula for the saturation of gain in a homogeneously-broadened laser system since gain is negative absorption.

[50] Here $I_\text{sat} = \hbar\omega A_{21}/(2\sigma(\omega_0))$ and $\sigma(\omega_0)$ is given by eqn 7.81.

We can also obtain eqn 7.82 directly from the steady-state value of $w = (N_1 - N_2)/N$ in eqn 7.68 if the saturation intensity is defined by

$$\frac{I}{I_{\rm sat}} = \frac{2\Omega^2}{\Gamma^2}\,. \tag{7.86}$$

This is equivalent to eqn 7.85.[51] At saturation the Rabi frequency has a value comparable with Γ.

[51] As discussed after eqn 7.75, Ω^2/I does not depend on the electric field.

7.6.3 Power broadening

Equation 7.84 for $\kappa(\omega, I)$ contains two quantities that vary with frequency: $\sigma(\omega)$ and $I_{\rm s}(\omega)$.[52] Rearranging this equation to show the frequency dependence, and defining $\sigma_0 \equiv \sigma(\omega_0)$ as the maximum cross-section in eqn 7.81, we find that

$$\begin{aligned}\kappa(\omega, I) &= N\sigma_0 \frac{\Gamma^2/4}{(\omega-\omega_0)^2 + \Gamma^2/4} \times \frac{1}{1 + \frac{I}{I_{\rm sat}}\frac{\Gamma^2/4}{(\omega-\omega_0)^2+\Gamma^2/4}} \\ &= N\sigma_0 \frac{\Gamma^2/4}{(\omega-\omega_0)^2 + \frac{1}{4}\Gamma^2(1 + I/I_{\rm sat})}\,. \end{aligned} \tag{7.87}$$

The expression for the absorption coefficient $\kappa(\omega, I)$ has a Lorentzian line shape with a full width at half maximum (FWHM) of

$$\Delta\omega_{\rm FWHM} = \Gamma\left(1 + \frac{I}{I_{\rm sat}}\right)^{1/2}\,. \tag{7.88}$$

The line width increases with intensity. This power broadening occurs because saturation reduces the absorption near the resonance while far from resonance the absorption changes little (see Fig. 7.7). The expression for the population in the upper level ρ_{22} in eqn 7.69 also has the same power-broadened line width, as in eqn 7.88.[53] The relationship between this absorption and the populations of the two levels is discussed in Exercise 7.11.

[52] In terms of the *minimum* value $I_{\rm sat} = I_{\rm s}(\omega_0)$ we can rewrite eqn 7.83 as

$$\frac{I_{\rm s}(\omega)}{I_{\rm sat}} = \frac{\sigma_0}{\sigma(\omega)}\,.$$

We also have

$$\sigma(\omega) = \sigma_0 \frac{\Gamma^2/4}{(\omega-\omega_0)^2 + \Gamma^2/4}\,.$$

[53] This can be shown by rearrangement of the denominator of eqn 7.69 in terms of a line width

$$\begin{aligned}\Delta\omega &= \Gamma\left(1 + \frac{2\Omega^2}{\Gamma^2}\right)^{1/2} \\ &= \Gamma\left(1 + \frac{I}{I_{\rm sat}}\right)^{1/2}\,.\end{aligned}$$

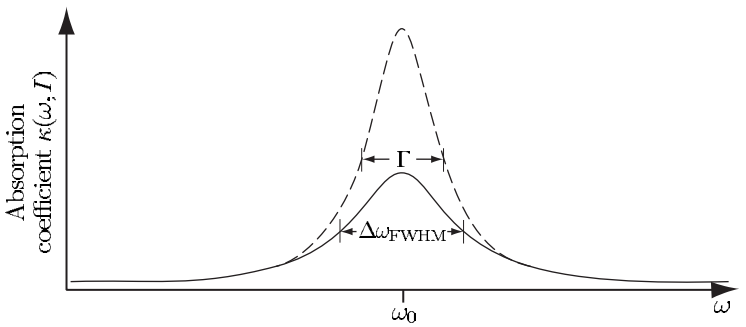


Fig. 7.7 The absorption coefficient $\kappa(\omega, I)$ is a Lorentzian function of the frequency that peaks at ω_0, the atomic resonance. Saturation causes the absorption line shape to change from the curve for a low intensity ($I \ll I_{\rm sat}$, dashed line), to a broader curve (solid line), with a lower peak value, described by the Lorentzian function in eqn 7.87.